```
import pandas as pd
```

```
data={
    "Student_Id":[101,102,103,104,105,106,107,108,109,110,110],
    "Name":["Anu","Bharat","Chetan","Divya","Esha","Farhan","Gowri","Harish",
            "Isha","Kiran","Kiran"],
    "Department":["CSE","AIML","CSE","ISE","AIML","CSE","ISE","CSE","AIML",
                  "ISE","ISE"],
    "Gender":["F","M","M","F","F","M","F","M","F","M","M"],
    "Attendence":[93.5,85.6,83.7,92.8,95.9,98.4,85.3,87.2,97.3,88.2,88.2],
    "Python_marks":[88,87,98,96,95,None,65,87,87,90,90],
    "Data_strcture_marks":[87,87,85,56,None,76,87,76,89,67,67],
    "Final_score":[85,88,89,67,68,90,95,None,98,95,95]

}
df=pd.DataFrame(data)
print(df)
```

```
    Student_Id    Name Department Gender  Attendence  Python_marks  \
0          101     Anu        CSE      F        93.5          88.0
1          102  Bharat       AIML      M        85.6          87.0
2          103  Chetan        CSE      M        83.7          98.0
3          104   Divya        ISE      F        92.8          96.0
4          105    Esha       AIML      F        95.9          95.0
5          106  Farhan        CSE      M        98.4           NaN
6          107   Gowri        ISE      F        85.3          65.0
7          108  Harish        CSE      M        87.2          87.0
8          109    Isha       AIML      F        97.3          87.0
9          110   Kiran        ISE      M        88.2          90.0
10         110   Kiran        ISE      M        88.2          90.0

    Data_strcture_marks  Final_score
0                  87.0         85.0
1                  87.0         88.0
2                  85.0         89.0
3                  56.0         67.0
4                   NaN         68.0
5                  76.0         90.0
6                  87.0         95.0
7                  76.0          NaN
8                  89.0         98.0
9                  67.0         95.0
10                 67.0         95.0
```

```
df.head()
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
0	101	Anu	CSE	F	93.5	88.0	87.0	85.0
1	102	Bharat	AIML	M	85.6	87.0	87.0	88.0
2	103	Chetan	CSE	M	83.7	98.0	85.0	89.0
3	104	Divya	ISE	F	92.8	96.0	56.0	67.0
4	105	Esha	AIML	F	95.9	95.0	NaN	68.0

```
df.head(3)
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
0	101	Anu	CSE	F	93.5	88.0	87.0	85.0
1	102	Bharat	AIML	M	85.6	87.0	87.0	88.0
2	103	Chetan	CSE	M	83.7	98.0	85.0	89.0

```
df.head(7)
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
0	101	Anu	CSE	F	93.5	88.0	87.0	85.0
1	102	Bharat	AIML	M	85.6	87.0	87.0	88.0
2	103	Chetan	CSE	M	83.7	98.0	85.0	89.0
3	104	Divya	ISE	F	92.8	96.0	56.0	67.0
4	105	Esha	AIML	F	95.9	95.0	NaN	68.0
5	106	Farhan	CSE	M	98.4	NaN	76.0	90.0
6	107	Gowri	ISE	F	85.3	65.0	87.0	95.0

```
df.tail()
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
6	107	Gowri	ISE	F	85.3	65.0	87.0	95.0
7	108	Harish	CSE	M	87.2	87.0	76.0	NaN
8	109	Isha	AIML	F	97.3	87.0	89.0	98.0
9	110	Kiran	ISE	M	88.2	90.0	67.0	95.0
10	110	Kiran	ISE	M	88.2	90.0	67.0	95.0

```
df.tail(3)
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
8	109	Isha	AIML	F	97.3	87.0	89.0	98.0
9	110	Kiran	ISE	M	88.2	90.0	67.0	95.0
10	110	Kiran	ISE	M	88.2	90.0	67.0	95.0

```
print(df.shape)
```

```
(11, 8)
```

```
print(df.columns.tolist())
```

```
['Student_Id', 'Name', 'Department', 'Gender', 'Attendence', 'Python_marks', 'Data_strcture_marks', 'Final_score']
```

Double-click (or enter) to edit

```
print(df.columns.tolist())
```

```
['Student_Id', 'Name', 'Department', 'Gender', 'Attendence', 'Python_marks', 'Data_strcture_marks', 'Final_score']
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 8 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Student_Id           11 non-null     int64  
 1   Name                 11 non-null     object 
 2   Department           11 non-null     object 
 3   Gender               11 non-null     object 
 4   Attendence           11 non-null     float64
 5   Python_marks         10 non-null     float64
 6   Data_strcture_marks  10 non-null     float64
 7   Final_score          10 non-null     float64
dtypes: float64(4), int64(1), object(3)
memory usage: 836.0+ bytes
```

```
print(df.dtypes)
```

```
Student_Id              int64
Name                   object
Department             object
```

```
Gender                 object
Attendence            float64
Python_marks          float64
Data_strcture_marks   float64
Final_score           float64
dtype: object
```

```
df.describe()
```

	Student_Id	Attendence	Python_marks	Data_strcture_marks	Final_score
count	11.000000	11.000000	10.000000	10.0000	10.000000
mean	105.909091	90.554545	88.300000	77.7000	87.000000
std	3.176619	5.204107	9.141481	11.2847	11.015141
min	101.000000	83.700000	65.000000	56.0000	67.000000
25%	103.500000	86.400000	87.000000	69.2500	85.750000
50%	106.000000	88.200000	89.000000	80.5000	89.500000
75%	108.500000	94.700000	93.750000	87.0000	95.000000
max	110.000000	98.400000	98.000000	89.0000	98.000000

```
print("mean of python marks:",df["Python_marks"].mean())
print("mean of data strcture marks:",df["Data_strcture_marks"].mean())
print("mode of python marks:",df["Python_marks"].mode())
```

```
mean of python marks: 88.3
mean of data strcture marks: 77.7
mode of python marks: 0    87.0
Name: Python_marks, dtype: float64
```

```
print("median of python marks:",df["Python_marks"].median())
print("median of data strcture marks:",df["Data_strcture_marks"].median())
```

```
median of python marks: 89.0
median of data strcture marks: 80.5
```

```
print("mode of python marks:",df["Python_marks"].mode())
```

```
mode of python marks: 0    87.0
Name: Python_marks, dtype: float64
```

```
print("\nMode of department")
print(df["Department"].mode())
print("\nMode of Gender")
print(df["Gender"].mode())
```

```

Mode of department
0    CSE
1    ISE
Name: Department, dtype: object

Mode of Gender
0    M
Name: Gender, dtype: object
```

```
df.describe()
```

	Student_Id	Attendence	Python_marks	Data_strcture_marks	Final_score
count	11.000000	11.000000	10.000000	10.0000	10.000000
mean	105.909091	90.554545	88.300000	77.7000	87.000000
std	3.176619	5.204107	9.141481	11.2847	11.015141
min	101.000000	83.700000	65.000000	56.0000	67.000000
25%	103.500000	86.400000	87.000000	69.2500	85.750000
50%	106.000000	88.200000	89.000000	80.5000	89.500000
75%	108.500000	94.700000	93.750000	87.0000	95.000000
max	110.000000	98.400000	98.000000	89.0000	98.000000

```
print("python marks")
print("Mean:",df["Python_marks"].mean())
print("Median:",df["Python_marks"].median())
print("Mode:",df["Python_marks"].mode())
print("Minimum:",df["Python_marks"].min())
print("Maximum:",df["Python_marks"].max())
print("Standard deviation:",df["Python_marks"].std())
```

```
python marks
Mean: 88.3
Median: 89.0
Mode: 0    87.0
Name: Python_marks, dtype: float64
Minimum: 65.0
Maximum: 98.0
Standard deviation: 9.141480551128831
```

```
df.isnull()
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	True	False
5	False	False	False	False	False	True	False	False
6	False	False	False	False	False	False	False	False
7	False	False	False	False	False	False	False	True
8	False	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False	False
10	False	False	False	False	False	False	False	False

```
print(df.isnull().sum())
```

```
Student_Id             0
Name                   0
Department             0
Gender                 0
Attendence             0
Python_marks           1
Data_strcture_marks    1
Final_score            1
dtype: int64
```

```
(df.isnull().sum()/len(df))*100
```

	0
Student_Id	0.000000
Name	0.000000
Department	0.000000
Gender	0.000000
Attendence	0.000000
Python_marks	9.090909
Data_strcture_marks	9.090909
Final_score	9.090909

```
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	False
10	True

dtype: bool

```
print("Number of duplicated rows:",df.duplicated().sum())
df.duplicated().sum()
```

```
Number of duplicated rows: 1
np.int64(1)
```

```
df[df.duplicated()]
```

	Student_Id	Name	Department	Gender	Attendence	Python_marks	Data_strcture_marks	Final_score
10	110	Kiran	ISE	M	88.2	90.0	67.0	95.0

```
Start coding or generate with AI.
```